How we calculate epcr (MVA Hackathon Track 1)

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What epcr is

epcr is the estimated probability that the causal relationship between a variant and the child's phenotype is correct. Our lead, the BUB1B pair, scores 0.72. We compute every value with a script (`submission/epcr.py`) instead of assigning it by hand, so each input can be checked and changed.

A variant is only causal if three things are all true, so epcr multiplies three probabilities:

epcr = P_variant × P_gene × P_inheritance

- **P_variant**: is the variant itself pathogenic?
- **P_gene**: does this gene cause this phenotype?
- **P_inheritance**: is the inheritance requirement met in this child (two hits for a recessive gene, penetrance for a dominant one)?

1. P_variant: is the variant pathogenic?

We score each variant with the standard ACMG/AMP criteria, then convert them to a probability using the published Bayesian point system (Tavtigian et al. 2018, *Genet Med* 20:1054; Tavtigian et al. 2020, *Hum Mutat* 41:1734). Each criterion adds points by strength:

Evidence strength	Example criteria	Points	Odds multiplier
Very strong	PVS1 (null variant in a loss-of-function gene)	+8	×350
Strong	PS1–PS4	+4	×18.7
Moderate	PM1 (critical domain)	+2	×4.33
Supporting	PP3 (predictors agree), PP4 (phenotype fits), PM2_Supporting (absent/rare in gnomAD)	+1	×2.08

Evidence strength	Example criteria	Points	Odds multiplier
Benign supporting	BP criteria	−1	÷2.08
Benign strong	BS criteria	−4	÷18.7

Starting from the framework's prior of 0.10 that a rare variant is pathogenic:

odds = (0.10 / 0.90) × 2.08^points, then **P_variant = odds / (1 + odds)**

This reproduces the familiar ACMG classes: 10+ points gives 0.99 or more (Pathogenic), 6–9 gives 0.90–0.99 (Likely pathogenic), 0–5 gives 0.10–0.81 (VUS). For a two-variant row, P_variant is the product of both, since both must be pathogenic.

2. P_gene: does this gene cause this phenotype?

We use ClinGen-style gene–disease validity tiers, each mapped to a fixed probability:

Validity tier	P_gene	Example in our list
Definitive	1.0	BUB1B → mosaic variegated aneuploidy type 1
Strong	0.9	
Moderate	0.7	
Limited	0.3	
No established association	0.02	DDX28, TRDMT1

We deliberately do not use Exomiser's combined score here. It ranks candidates well, but it is not a calibrated probability that the gene is this patient's diagnosis.

3. P_inheritance: is the inheritance requirement met?

This factor depends on the gene's inheritance mode and what we see in the child:

Situation	P_inheritance	Reasoning
Recessive gene, two variants, phase unknown (BUB1B)	0.9	Two rare alleles start at 0.5 for being on different copies; a phenotype matching the gene raises that sharply, because if both were on one copy the child would be an unaffected carrier
Recessive gene, one variant, primary candidate	0.05	Chance of a second hit our data missed (deep intronic or structural)
Recessive gene, one variant, reported as carrier (secondary)	0.02	Carrier state only
Dominant gene, primary candidate	1.0	One hit is enough
Dominant gene, secondary finding	penetrance	LZTR1 schwannomatosis 0.4; one FLG null allele 0.15

We do not also score ACMG criterion PM3 ("in trans with a pathogenic variant"), because that would count phase twice.

Phase evidence from the sequencer is used where it exists: the two FANCD2 deletions share a GATK phase set, which proves they are on the same copy, so FANCD2 is scored as a single hit.

Worked example: BUB1B scores 0.72

The two BUB1B variants are scored separately, then combined:

Variant	ACMG criteria	Points	P_variant
c.2210T>G p.Leu737* (stop codon, ClinVar pathogenic)	PVS1 +8, PM2_Supporting +1	9	0.988
c.3006T>G p.Asn1002Lys (kinase domain, absent from gnomAD)	PM2_Supporting +1, PM1 +2, PP3 +1, PP4 +1	5	0.812

epcr = (0.988 × 0.812) × 1.0 × 0.9 = 0.802 × 1.0 × 0.9 = 0.72

The score is held back mostly by the novel missense, which is a strong VUS rather than likely pathogenic under strict ACMG rules. Proving the two variants are on different copies (parental testing or long reads) would raise the pair to about 0.80; a functional assay on Asn1002Lys would push it above 0.9.

All 10 submitted rows

Only BUB1B is a plausible diagnosis; the other primary rows are low-probability alternatives, and the secondary rows are incidental findings scored by penetrance.

Gene	Variant	Type	Variant evidence (points)	P_variant	P_gene	P_inheritance	epcr
BUB1B	p.Leu737* + p.Asn1002Lys	primary	9 and 5 (see above)	0.802	1.0	0.9	**0.722**
CACNB1	stop-lost	primary	PM2, PM4 (3)	0.500	0.02	1.0	0.010
DDX28	p.Arg436Trp	primary	PM2, PP3 (2)	0.325	0.02	1.0	0.006
TRDMT1	p.Leu377Phe	primary	PM2, PP3 (2)	0.325	0.02	1.0	0.006
SLC34A1	p.Ala133Val	primary	BS1, PP3 (−3)	0.012	0.3	1.0	0.004
MAD1L1	p.Arg59Cys	primary	none (0)	0.100	0.3	0.05	0.002
FANCD2	c.1278+1del	primary	BS (−4)	0.006	1.0	0.05	0.001
LZTR1	p.Tyr748*	secondary	PVS1, PM2 (9)	0.988	1.0	0.4	0.395
FLG	p.Arg2447*	secondary	PVS1 (8)	0.975	1.0	0.15	0.146
GNRHR	p.Arg139His	secondary	PS3, PM2, PP3 (6)	0.900	1.0	0.02	0.018

Every factor for every row is listed in `submission/epcr_audit.tsv`, which the script writes alongside the CSV.

Assumptions and what we may tweak

The method doesn't remove judgement; it moves it into named decisions anyone can check one at a time. These are the least settled:

Assumption	Current value	Effect on BUB1B if changed
PM1 (critical kinase domain) applies to Asn1002Lys	+2 points	Dropping it lowers BUB1B to about 0.44
BUB1B variants are on different copies	0.9	0.5 (no update for phenotype) gives 0.40
LZTR1 schwannomatosis penetrance	0.4	Secondary row only
FLG null-allele penetrance	0.15	Secondary row only
Prior that a rare variant is pathogenic	0.10 (framework default)	Kept as published

All values are constants in `submission/epcr.py`, each with a comment; ones marked `TWEAK` are the ones above. Running the script rewrites the `epcr` column of `submission/track1_submission.csv` and the audit table. The full write-up is in `submission/EPCR_METHOD.md`.